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1 / 2 submitted

Q1. Smart Hospital Patient Registration System

Score: 0.00 TC: 0/3 passed

CODE

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<String[]> patientList = new ArrayList<>();
        HashMap<String, String[]> patientMap = new HashMap<>();
        HashMap<String, Integer> deptCount = new HashMap<>();

        int choice;

        do {
            choice = Integer.parseInt(sc.nextLine().trim());

            switch (choice) {
                case 1: {
                    String id = sc.nextLine().trim();

                    if (patientMap.containsKey(id)) {
                        System.out.println("Duplicate Patient ID.");
                        break;
                    }

                    String name = sc.nextLine().trim();
                    if (!name.replaceAll("\\s", "").matches("[A-Za-z]+")) {
                        System.out.println("Invalid Name.");
                        break;
                    }

                    String mobile = sc.nextLine().trim();
                    if (!mobile.matches("\\d{10}")) {
                        System.out.println("Invalid Mobile Number.");
                        break;
                    }

                    String dept = sc.nextLine().trim().toUpperCase();
                    String doctor = sc.nextLine().trim();

                    String properName = toProperCase(name);

                    String[] patient = {id, properName, mobile, dept, doctor};

                    patientList.add(patient);
                    patientMap.put(id, patient);

                    deptCount.put(dept, deptCount.getOrDefault(dept, 0) + 1);

                    System.out.println("Patient Registered Successfully.");
                    break;
                }

                case 2: {
                    String id = sc.nextLine().trim();
                    if (patientMap.containsKey(id)) {
                        String[] p = patientMap.get(id);
                        System.out.println("Patient Details");
                        System.out.println("ID : " + p[0]);
                        System.out.println("Name : " + p[1]);
                        System.out.println("Mobile : " + p[2]);
                    }
                }
            }
        } while (choice != 0);
    }
}
```

```
        System.out.println("Department : " + p[3]);
        System.out.println("Doctor : " + p[4]);
    } else {
        System.out.println("Patient Not Found");
    }
    break;
}

case 3: {
    System.out.println("-----");
    System.out.println("ID Name Department");
    System.out.println("-----");
    for (String[] p : patientList) {
        System.out.println(p[0] + " " + p[1] + " " + p[3]);
    }
    break;
}

case 4: {
    System.out.println("Department Count");
    for (Map.Entry<String, Integer> entry : deptCount.entrySet()) {
        System.out.println(entry.getKey() + " : " + entry.getValue());
    }
    break;
}

case 5: {
    break;
}

default: {
    System.out.println("Invalid Choice.");
}
}

} while (choice != 5);

sc.close();
}

public static String toProperCase(String str) {
    String[] words = str.trim().split("\\s+");
    StringBuilder result = new StringBuilder();

    for (String word : words) {
        if (word.length() > 0) {
            result.append(Character.toUpperCase(word.charAt(0)))
                .append(word.substring(1).toLowerCase())
                .append(" ");
        }
    }

    return result.toString().trim();
}
```

OUTPUT

Q2. Online Library Book Management System

— Not Submitted —